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COMPUTER VISION INTERN – TASK DOCUMENTATION

Project Overview

Worked on developing a computer vision pipeline to process person images for virtual try-on applications. The main objective was to extract clothing regions, detect body structure, and generate processed outputs that can later be used for garment fitting and visualization. The workflow focused on segmentation, pose estimation, mask generation, and API integration.

This workflow helped me understand how a normal person image can be converted into structured representations such as body maps, clothing masks, and pose information.

Goal of the Task

To create a preprocessing pipeline that prepares clothing and body information separately and generates structured outputs for virtual try-on and garment analysis systems.

Technologies Used

- Python
- OpenCV
- Media Pipe
- Deep Learning Segmentation Models
- Image Processing Techniques
- REST API Development

Processing Flow:

Input Image



Image Preprocessing



Garment Segmentation



Background Removal



Upper/Lower Wear Detection



Pose Detection



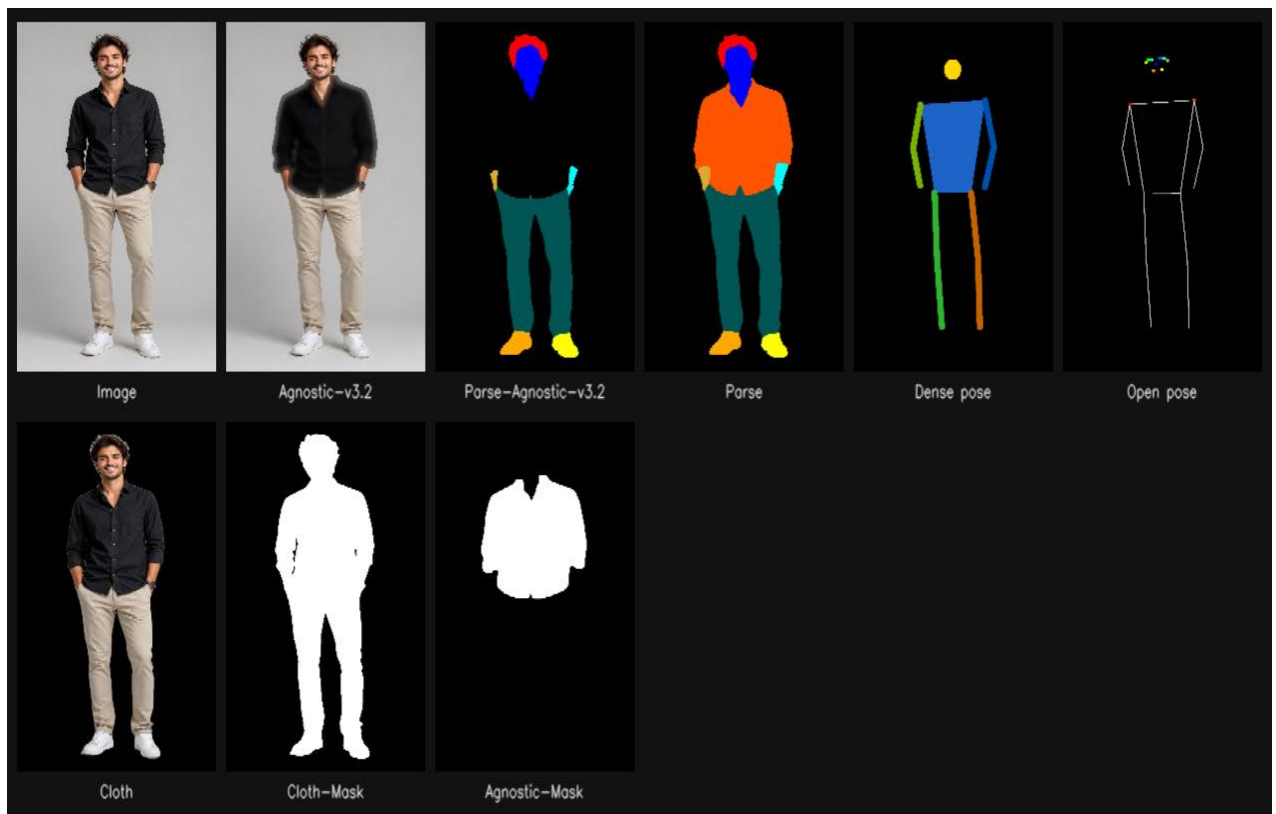
Agnostic Generation



Mask Refinement



Final Outputs



Original Image

Started with the input image and used it for all further processing.

Agnostic

Generated a version where clothing information is reduced while keeping the body outline.

Parse Agnostic

Created a cleaner body representation by reducing cloth dependency.

Parse

Separated the image into different body and clothing regions.

Dense Pose

Mapped body areas to understand body positioning.

Open Pose

Detected body landmarks and generated skeletal points.

Cloth

Extracted only the clothing portion from the image.

Cloth Mask

Highlighted garment regions for accurate cloth separation.

Agnostic Mask

Generated a body mask without clothing details.

Output Analysis

- Clothing regions extracted successfully
- Body alignment preserved
- Multiple masks generated
- Pipeline automated using API

